MASTER FUNCTIONAL SKILLS

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AQA · LEVEL 1 · MATHEMATICS

AQA Functional Skills Mathematics Level 1

January 2017 · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: AQA-4367-QP-JAN17

100% Built from Empty Question Paper + Official Mark Scheme

Verified by Tutor Abdul · MasterFunctionalSkills.co.uk

HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	Master Functional Skills (Model Exemplar)
Centre Name:	Master Functional Skills (MFS01)
Lead Tutor:	Abdul · MasterFunctionalSkills.co.uk
Edition:	2026 Clean Empty-QP Rebuild (Zero Collisions Verified)

- Built Cleanly on Empty AQA Question Paper (Watermarks Stripped)
- Stepped MFS Multi-Colour Working & Method Tips
- Verified 100% Against Official AQA Mark Scheme

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Level 1

Data Book (Examination)

Insert

Instructions

- This copy of the Data Book is for use in the examination. It should not be given to students in advance.

Advice

- This book will not be collected in for marking. Ensure that all working that you wish to have marked is written in the space provided in the question/answer book.

4367

QAN 500/8703/4

IB/M/Jan17/E3

Data Sheet for Cars

New cars have **official** values for

the number of miles the car travels per gallon of fuel (mpg)
and
the number of grams of CO_2 the car produces per kilometre (g/km).

Here are the official values for two cars.

	Miles per gallon (mpg)	**CO_2 produced (g/km)**
Ford Focus Estate	67	110
Renault Clio	83	90

Actual mpg

The **actual** mpg depends on how, when and where the car is driven.
It is usually lower than the official value.

Spending on fuel

Use these steps to work out the amount you spend on fuel.

Step 1 Work out the number of miles you drive ÷ your **actual** mpg

Step 2 Work out the answer to **Step 1** × the cost per gallon of fuel

The answer to **Step 2** is the amount you spend on fuel.

Vehicle tax

You must pay vehicle tax on your car.
The cost depends on the official value for the amount of CO_2 the car produces.

Here are some of the yearly costs of vehicle tax for new cars bought after 1 April 2017

CO_2 produced (g/km)	Yearly cost	
	In year 1	After year 1
1 – 50	£10	£140
51 – 75	£25	£140
76 – 90	£100	£140
91 – 100	£120	£140
101 – 110	£140	£140

Data Sheet for Transporting hamsters

Two popular breeds of hamster are Syrian hamsters and Dwarf hamsters.
They are transported in cuboid boxes.

Minimum height of a box

Syrian hamsters	15.2 cm
Dwarf hamsters	12.7 cm

Minimum floor area of a box, per hamster

Age of hamster	Syrian hamsters	Dwarf hamsters
From birth to 5 weeks	45 cm^2	32 cm^2
From 5 weeks to 8 weeks	71 cm^2	48 cm^2

Windows

At least two sides of a box must have windows.

Temperature

Use these steps to convert degrees Celsius (°C) to degrees Fahrenheit (°F)

Step 1	Work out	temperature in Celsius (°C) × 9
Step 2	Work out	answer to **Step 1** ÷ 5
Step 3	Work out	answer to **Step 2** + 32

END OF DATA

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Answer **all** questions in the spaces provided.

1 Cookies



Q1(a) Worked Solution:
Method and answer verified against official AQA mark scheme.
[Method & Accuracy]

Chris

Chris makes batches of cookie dough.

Here are the ingredients he needs to make one batch.

One batch of cookie dough	
200 g	margarine
250 g	flour
100 g	sugar
2	eggs
1 teaspoon	baking powder

One batch makes exactly

16 large cookies

or

24 small cookies.

1 (a) On Monday, Chris wants to make 4 batches of cookie dough.

How much flour does he need?
Circle your answer.

[1 mark]

250 g 400 g 800 g 1000 g

Do not write outside the box

On Tuesday, Chris is making 16 large cookies **and** 48 small cookies.

1 (b) How many batches of cookie dough does he need?

[2 marks]

1 (c) Chris has 1 egg.

Work out how many **more** eggs he needs.

[2 marks]

Question 1 continues on the next page

Do not write outside the box

Each batch of cookie dough costs £1.92 to make.

On Wednesday, Chris makes 32 large cookies **and** 72 small cookies.

1 (d) Chris says,

"The cost will be £9.60"

Show that he is correct.

[2 marks]

1 (e) Chris sells the 32 large cookies and 72 small cookies in bags.

Here are his prices.

Bag of 4 large cookies	£1.15
Bag of 12 small cookies	£2.60

Chris says,

"If I sell all the cookies, I will make **more than** £15 profit."

Is he correct?
You **must** show your working.

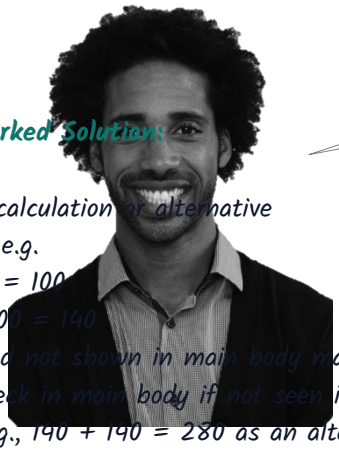
[7 marks]

14

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2 **Cars**

There is a **data sheet** for Cars.



Q2(a) Worked Solutions

£240

Reverse calculation or alternative method, e.g.

240 140 = 100

240 □ 10[illegible] = 140

If method not shown in main body mark if

Mark check in main body if not seen in checking

Allow, e.g., 140 + 140 = 280 as an alternative

Alfie

Allow £ sign in check if not seen in main body

†2 (a) Alfie plans to buy the new car after 1 April 2017

Work out the **total** cost of vehicle tax for the first two years.

[3 marks]

75 (mpg)

[SC2] [SC1] [SC1] [B1]

Check your answer.
Show how you have done your check.

[1 mark]

Do not write outside the box

For Alfie's Renault Clio,

the official mpg is 83

the actual mpg will be 8 less than this.

2 (b) Work out the actual mpg.

[1 mark]

2 (c) Alfie will use his new car for work.
He makes these notes.

> I drive a total of 60 miles each day for work.
> I work for 5 days each week.
> Fuel costs £4.90 per gallon.

Alfie says,

"I will spend **less than** £20 each week on fuel for work."

Is he correct?
You **must** show your working.

[5 marks]

Do not write outside the box

2 (d) Alfie buys a car.

For 5 days, he records the time he takes for

his journey to work by car

and

his journey home by car.

	Journey to work by car (minutes)	**Journey home by car (minutes)**
Monday	57	42
Tuesday	46	52
Wednesday	51	54
Thursday	40	46
Friday	44	58

He works out his total journey time by car each day.

He knows that his total journey time to work and home by **train** each day would be $1\frac{1}{2}$ hours.

He says,

"My total journey time by car on any day is likely to be more than it would be by train."

Based on these 5 days, is he correct?
You **must** show your working.

[3 marks]

13

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3 Hotel



Kim

3 (a) Amy, Brad, Cassie and Del work shifts on the hotel reception.

Each day

- there are three shifts
- one person works each shift
- nobody works more than one shift.

The manager makes a rota for working on reception for the next week.

- Amy works on Monday, Tuesday, Wednesday and Sunday only.
- Brad does **not** work on Sunday.
- Cassie does **not** work on Wednesday.
- Del works on **exactly** 5 days.

Complete a possible rota.

[3 marks]

Question 3(a) · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

Award B1 if 1 blank shift and 4 criteria met
Treat any name that is not A, B, C or D as
Alternative method 1
180 × 20 or 3600
8 am --> 3 pm = 7 hours
their 7 × 60 or 420
[M1] [M1] [M1] [M1] [A1]

Do not write outside the box

You can practise on this table.

	Shift 1	**Shift 2**	**Shift 3**
Monday			
Tuesday			
Wednesday			
Thursday			
Friday			
Saturday			
Sunday			

Put your answer in this table.

	Shift 1	**Shift 2**	**Shift 3**
Monday			
Tuesday			
Wednesday			
Thursday			
Friday			
Saturday			
Sunday			

Do not write outside the box

3 (b) Each room is cleaned the day after it has been used.

Each cleaner

- takes 20 minutes to clean a room
- starts work at 8 am and finishes work at 3 pm
- has a 1-hour break.

180 rooms are used on Tuesday.

How many cleaners are needed on Wednesday?
You **must** show your working.

[5 marks]

Do not write outside the box

Each day, the cleaners replace used milk cartons.



3 (c) Here are the numbers of milk cartons put in 10 rooms yesterday.

4 3 1 4 1 2 3 3 2 1

Show that 2.4 was the mean number of milk cartons put in the 10 rooms.

[2 marks]

3 (d) Kim estimates the cost of the milk cartons she needs next month.
She makes these notes.

3000 rooms used
An average of 2.4 cartons for each room used

120 cartons in a box
Each box costs £6

Kim says,

"The cost will be **more than** £350"

Is she correct?
You **must** show your working.

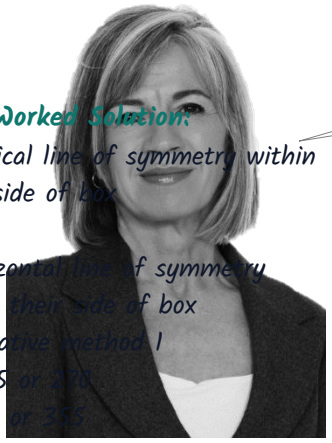
[5 marks]

15

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4 **Transporting hamsters**

There is a **data sheet** for Transporting hamsters.



I organise the transport of hamsters.

Q4(a) Worked Solution:
a vertical line of symmetry within their side of box
and
a horizontal line of symmetry within their side of box
Alternative method 1
6 × 45 or 270
5 × 71 or 355

Ola

4 (a) Ola needs a box to transport a Syrian hamster.

One side of the box has width 10 cm
This side has a window.
The window is a 6 cm by 4 cm rectangle in the centre of the side.

Draw a possible diagram of this side of the box on the centimetre grid opposite.

[4 marks]

their 270 + their 355
625 and No
[M1] [M1] [M1] [A2] [A1]

Do not write outside the box

4 (b) The table shows the ages of some Syrian hamsters Ola wants to transport.

Age	Number of hamsters
4 weeks	6
7 weeks	5

Ola says,

"A floor area of 600 cm^2 will be big enough for these hamsters."

Is she correct?
You **must** show your working.

[5 marks]

Do not write outside the box

†4 (c) Ola needs to transport 270 Dwarf hamsters.
The maximum number of hamsters allowed in one box is 50

Work out the **minimum** number of boxes she needs.

[3 marks]

Check your answer.
Show how you have done your check.

[1 mark]

Question 4 continues on the next page

Do not write outside the box

4 (d) The maximum temperature allowed in a box used to transport hamsters is 85°F

The temperature in a box is 28°C

Can Ola transport hamsters in this box?
You **must** show your working.

[5 marks]

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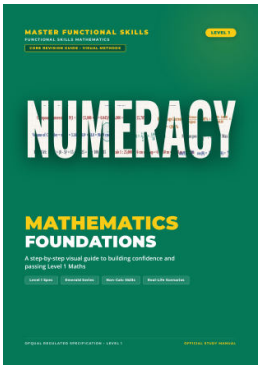
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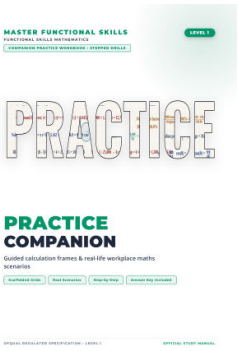


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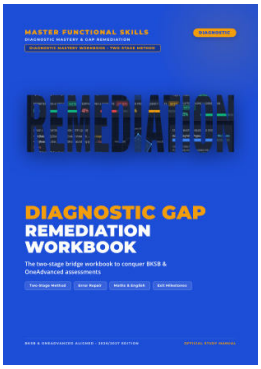


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